STAT3007/7007 Deep Learning

The schedule is tentative and will be subject to changes.

Week	Date	Activity	Title
			Course overview & machine learning basics
W1	29/07	L1 T1 P1	Introduction. Regression. Classification
W2	05/08	L2 T2 P2	PCA. Statistical learning. Model selection.
			Neural network basics
W3	**12/08** TBD	L3 T3 P3	Perceptron. Adaline. Hopfield Networks.
W4	19/08	L4 T4 P4	Gradient-based learning. MLPs. Automatic differentiation.
			Deep Architectures
W5	26/08	L5 T5 P5	Convolutional neural networks
W6	02/09	L6 T6 P6	Recurrent neural networks
	04/04		**A1 due**
			Optimization for deep learning
W7	09/09	L7 T7 P7	Optimization basics. Initialization and Input Transformation. Batch normalization. Adaptive learning rate.
			Improving Generalization
	14/04		**Project proposal due**
W8	16/09	L8 T8 P8	Regularization. Adversarial examples and adversarial training.
W9	23/09	L9 T9 P9	Activation functions. ResNet, HighwayNet, DenseNet. Attention.
			In-semester break (28/9 – 2/10)
			Generative Models
W10	07/10	L28 T10 P10	Autoencoders. Variational auto-encoder. Generative Adversarial Networks (GANs). Diffusion Models
			Reinforcement Learning
W11	14/10	L11 T11 P11	Reinforcement learning
	16/10		**A2 due**
			Miscellaneous Topics
W12	21/10	L12 T12 P12	Graph neural networks. Neural differential equation solvers. Discrete optimization.
W13	28/10	L13	Foundation models. Review. Project presentations

30/10	**Project slides due**
09/11	**Project report due**
10/11	**Project reflective essay due**